

# Liam Shannon

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## EDUCATION

### Queen's University

Expected May 2028

*BASc in Mechatronics and Robotics Engineering | Dean's Scholar | GPA: 4.19/4.3*

## SKILLS

**Languages:** TypeScript, JavaScript, Python, C/C++

**Tools & Frameworks:** ROS2, Claude Code, Ontology building, React, Git, Docker, Linux

**Data:** Python data pipelines, data cleaning & normalization, SQL-style querying/Foundry querying logic

**Other:** SolidWorks, Onshape, Altium, LTspice

## EXPERIENCE

### Autonomy Manager

July 2026 - Present

*Queen's Aerospace Design Team*

*Kingston, Canada*

- > Enabling autonomous SAE Aero mission execution by developing ROS 2 autonomy software integrated with PX4
- > Identifying task ground targets mid-flight from a monocular camera feed by developing a computer vision stack that pairs physical AI world models with a trained YOLO object-detection model
- > Directing a 15-member team building the software stack in C++ and Python for a fixed-wing autonomous drone to compete at the advanced class of the SAE Aero 2027 design competition
- > Managing a GitHub organization and a Dockerized Ubuntu 24.04 build system enabling consistent cross-platform development across macOS, Windows, and Linux for all contributors

### Software Developer

June 2026 - September 2026

*Cavtera*

*Ottawa, Canada*

- > Restored live equipment telemetry by fixing broken ingestion APIs and rebuilding the Python transform pipeline that parses 10,000+ records per day across a database of millions
- > Saved \$1,000,000+ annually in recoverable costs through exposing 10,000+ hours per week of idling equipment by building a full-stack equipment management system using TypeScript and React
- > Increased job-site utilization of 1,000+ trucks and pieces of equipment by shipping automated alerts
- > Modeled the business as a Foundry ontology and wrote the TypeScript edit functions behind it, giving non-technical staff a governed way to update production records
- > Shipped the operator-facing front end as a React and TypeScript application on Foundry's Ontology SDK (OSDK), so site staff could act on live fleet data directly

### Software Developer

May 2025 - April 2026

*Thomas Cavanagh Construction Limited*

*Ottawa, Canada / Remote*

- > Digitized dispatch for a 100+ driver fleet by building custom dispatch software
- > Eliminated 4+ hours of daily dispatcher calls and dozens of spreadsheets by modeling fleet data in an ontology
- > Reduced compute costs from \$100+/week to single digits by caching data locally instead of querying the ontology
- > Delivered the dispatcher-facing tooling as Foundry Workshop applications backed by Python transforms

## PROJECTS

### Autonomous Mars Rover Robot | ROS, Raspberry Pi 4, Arduino, SLAM

January 2026 - April 2026

- > Modeled the rover's kinematics, as measured by a 7-link/6-joint URDF spanning chassis, four wheel joints, and LiDAR, by authoring a modular Xacro description for Gazebo/RViz simulation
- > Achieved reliable Pi-Arduino motor control, as measured by a 100 Hz encoder loop over 115200-baud serial, by writing a custom serial protocol and ROS 2 node driving motors from `/cmd_vel`
- > Delivered real-time mapping and localization, as measured by continuous occupancy-grid generation from 12m-range LiDAR scans, by configuring a SLAM Toolbox pipeline on the robot's TF/odometry stack
- > Enabled remote teleoperation of the rover, as measured by a 20 Hz joystick input rate with deadman-switch safety, by building a ROS 2 teleop\_twist\_joy pipeline that drives `/cmd_vel` into the motor serial node

### Line Following Robot | Arduino, C++, Closed-Loop Control

January 2025 - April 2025

- > Developed a line-following robot using an Arduino to program motor actions based on closed-loop sensor feedback
- > Tuned algorithms based on sensor data and optimized performance by integrating hardware and software subsystems